Control work 5 — angles

Angles on a line, at a point, in triangles and in quadrilaterals.

LESSON 87 | 1 × 40 MIN | MATEMATIKA 6, NAZORAT ISHI 5 | S7 5 REVIEW

LESSON CLOCK

2' | 28' | 8' | 2'

Instructions	2 min
The paper	28 min
Answers and diagnosis	8 min
What comes next	2 min

LEARNING OBJECTIVES

- Classify and measure angles accurately.
- Use the four angle facts with a reason each time.
- Find angles in triangles and quadrilaterals.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 223–246
Cambridge	Stage 7, pp. 55–62
Workbook	Control paper 5

01 Explanation

The paper — 20 marks, 28 minutes

Q	TASK	MARKS	FROM
1	Classify $37°$, $90°$, $128°$ and $245°$	2	L80
2	x and $118°$ on a straight line: find x with a reason	3	L81–83
3	x, $3x$ and $140°$ at a point: find x	3	L81–83
4	$5x$ and $2x + 60$ vertically opposite: find the angle	3	L81–83
5	A triangle with $72°$ and $43°$: find the third	3	L84–86
6	An isosceles triangle with apex $34°$: find the base angles	3	L84–86
7	A quadrilateral with $85°$, $95°$ and $110°$: find the fourth	3	L84–86

THE ANSWERS

acute, right, obtuse, reflex; $62°$; $55°$; $100°$; $65°$; $73°$ each; $70°$.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
reflex called obtuse	$245°$	over $180°$ is reflex
no reason given	$x = 62°$	“angles on a straight line”
angles at a point summed to $180°$	$x + 3x + 140 = 180$	$x + 3x + 140 = 360$
vertically opposite set equal to $180°$	$5x + 2x + 60 = 180$	$5x = 2x + 60$
angle sum taken as $360°$ for a triangle	$360 - 115$	$180 - 115$
apex used as a base angle	$34°, 34°, 112°$	$34°, 73°, 73°$
quadrilateral summed to $180°$	a negative answer	$360°$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

IF YOU LOST MARKS ON	IT RETURNS IN
Q1	every diagram for the rest of the year
Q2–Q4	the circle, where angles at the centre fill $360°$
Q5–Q6	Grade 7, where these facts are proved
Q7	the pie chart in Quarter IV

LOOKING FORWARD

The rest of Quarter II is the circle: its parts, and the length of the curve round it. The $360°$ of this paper is what makes a pie chart possible.

02 Worked examples

EXAMPLE 1 Model answer, Q3: x, $3x$ and $140°$ at a point.

1. $x + 3x + 140 = 360$

   Angles at a point.

2. $4x = 220$

3. $x = 55°$

   Check: $55 + 165 + 140 = 360$ ✓

ANSWER

$55°$

EXAMPLE 2 Model answer, Q4: $5x$ and $2x + 60$ vertically opposite.

1. Vertically opposite angles are equal.

2. $5x = 2x + 60$

3. $3x = 60$, so $x = 20$.

4. Each angle is $100°$.

ANSWER

$100°$

EXAMPLE 3 **Model answer, Q6: isosceles with apex $34°$.**

1. $180 - 34 = 146$ for the two base angles.
2. $146 \div 2 = 73$.
3. $73°$ each.

   Check: $34 + 73 + 73 = 180$ ✓

ANSWER

$73°$ each

03 Interactive model

Mark Q2 for the reason alone; the class discovers that the number without a reason is worth less than half the marks.

The chapter in eight questions

$245°$ is:

- acute
- obtuse
- straight
- reflex

x with $118°$ on a line:

- $62°$
- $72°$
- $242°$
- $118°$

Its reason:

- angles at a point
- angles on a straight line
- vertically opposite
- triangle sum

$x + 3x + 140 = \ldots$

- 180
- 360
- 90
- 270

So x is:

- $10°$
- $55°$
- $40°$
- $80°$

$5x$ and $2x + 60$ vertically opposite give the angle:

- $20°$
- $60°$
- $100°$
- $180°$

A triangle with $72°$ and $43°$:

- $65°$
- $75°$
- $115°$
- $245°$

A quadrilateral with $85°$, $95°$, $110°$:

- $60°$
- $70°$
- $80°$
- $90°$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Acute angle	O‘tkir burchak	Острый угол
Obtuse angle	O‘tmas burchak	Тупой угол
Vertically opposite	Vertikal burchaklar	Вертикальные углы
Angle sum	Burchaklar yig‘indisi	Сумма углов
Isosceles	Teng yonli	Равнобедренный
Reason	Asos	Обоснование
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Q1 tests:

- calculating
- classifying
- measuring
- drawing

Q2 needs, besides the number:

- a diagram
- a reason
- a measurement
- nothing

Q3 sums to:

- $90°$
- $180°$
- $360°$
- $540°$

Q4 sets the two expressions:

- equal
- to $180°$
- to $360°$
- to $90°$

Q6 divides by:

- 2
- 3
- 4
- nothing

Q7 sums to:

- $180°$
- $270°$
- $360°$
- $540°$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Classify $37°$

   ANSWER Acute

2. Classify $128°$

   ANSWER Obtuse

3. Classify $245°$

   ANSWER Reflex

4. x with $118°$ on a line

   ANSWER $62°$

5. x, $3x$, $140°$ at a point

   ANSWER $x = 55°$

6. A triangle with $72°$ and $43°$

   ANSWER $65°$

7. A quadrilateral with $85°$, $95°$, $110°$

   ANSWER $70°$

Medium · the standard question

7 PROBLEMS

1. $5x$ and $2x + 60$ vertically opposite

   ANSWER $100°$

2. Isosceles with apex $34°$

   ANSWER $73°$ each

3. Isosceles with base angle $34°$

   ANSWER Apex $112°$

4. Two supplementary angles in the ratio $3 : 7$

   ANSWER $54°$ and $126°$

5. An angle twice its complement

   ANSWER $60°$

6. A parallelogram with one angle $115°$

   ANSWER $115°, 65°, 115°, 65°$

7. The reflex angle beside $118°$

   ANSWER $242°$

Hard · stretch and reason

7 PROBLEMS

1. A triangle with angles x, $x + 20$, $x + 40$

   ANSWER $40°$, $60°$, $80°$

2. A quadrilateral with angles x, $2x$, $3x$, $4x$

   ANSWER $36°$, $72°$, $108°$, $144°$

3. Four angles at a point, three equal and one $120°$

   ANSWER $80°$ each

4. An isosceles triangle with one angle of $90°$

   ANSWER $90°$, $45°$, $45°$

5. A triangle cannot have two obtuse angles: why?

   ANSWER They already exceed $180°$

6. An exterior angle of a triangle beside $65°$

   ANSWER $115°$

7. Why is a reason needed for every angle answer?

   ANSWER The diagram is not evidence; the fact is

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on, with its reason.

1. Rewrite in full every question on which you lost a mark.
2. Find x if x and $143°$ lie on a straight line, giving a reason.
3. Find x if $2x$, $4x$ and $120°$ meet at a point.
4. An isosceles triangle has an apex of $50°$. Find the base angles.
5. A quadrilateral has angles $100°$, $75°$ and $95°$. Find the fourth.